

Task 1: Create a Security Group for the RDS DB Instance

In this task, you will create a security group to allow your web server to access your RDS DB instance. The security group will be used when you launch the database instance.

4. In the AWS Management Console, in the search box next to **Services**, search for and select **VPC**.
5. In the left navigation pane, choose **Security groups**.
6. Choose **Create security group** and then configure:
 - **Security group name:** DB Security Group
 - **Description:** Permit access from Web Security Group
 - **VPC:** Lab VPC

Tip: Choose the X next to VPC that is already selected, then choose **Lab VPC** from the menu.

- 7.
8. In the **Inbound rules** pane, choose **Add rule**

The security group currently has no rules. You will add a rule to permit access from the *Web Security Group*.

9. Configure the following settings:
 - **Type:** MySQL/Aurora (3306)
 - **Source:** Place your cursor in the field to the right of Custom, type **sg**, and then select *Web Security Group*.

This configures the Database security group to permit inbound traffic on port 3306 from any EC2 instance that is associated with the *Web Security Group*.

10. Choose **Create security group**

You will use this security group when launching an Amazon RDS database in this lab.

Task 2: Create a DB Subnet Group

In this task, you will create a *DB subnet group* that is used to tell RDS which subnets can be used for the database. Each DB subnet group requires subnets in at least two Availability Zones.

10. In the AWS Management Console, in the search box next to **Services**, search for and select **RDS**.

11. In the left navigation pane, choose **Subnet groups**.

If the navigation pane is not visible, choose the menu icon in the top-left corner.

12. Choose **Create DB Subnet Group** then configure:

- **Name:** DB-Subnet-Group
- **Description:** DB Subnet Group
- **VPC:** Lab VPC

13. Scroll down to the **Add subnets** section.

14. Expand the list of values under **Availability Zones** and select the first two zones: **us-east-1a** and **us-east-1b**.

15. Expand the list of values under **Subnets** and select the subnets associated with the CIDR ranges **10.0.1.0/24** and **10.0.3.0/24**.

These subnets should now be shown in the **Subnets selected** table.

16. Choose **Create**

You will use this DB subnet group when creating the database in the next task.

Task 3: Create an Amazon RDS DB Instance

In this task, you will configure and launch a Multi-AZ Amazon RDS deployment of a MySQL database instance.

Amazon RDS **Multi-AZ** deployments provide enhanced availability and durability for Database (DB) instances, making them a natural fit for production database workloads. When you provision a Multi-AZ DB instance, Amazon RDS automatically creates a primary DB instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ).

17. In the left navigation pane, choose **Databases**.

18. Choose **Create database**

If you see **Switch to the new database creation flow** at the top of the screen, please choose it.

19. Select **MySQL** under **Engine Options**.

20. Under **Templates** choose **Dev/Test**.

21. Under **Availability and durability** choose **Multi-AZ DB instance**.

22. Under **Settings**, configure:

- **DB instance identifier:** lab-db
- **Master username:** main
- **Master password:** lab-password
- **Confirm password:** lab-password

23. Under **DB instance class**, configure:

- Select **Burstable classes (includes t classes)**.
- Select *db.t3.micro*

24. Under **Storage**, configure:

- **Storage type:** *General Purpose (SSD)*
- **Allocated storage:** 20

25. Under **Connectivity**, configure:

- **Virtual Private Cloud (VPC):** *Lab VPC*

26. Under **Existing VPC security groups**, from the dropdown list:

- Choose *DB Security Group*.
- Deselect *default*.

27. Under **Monitoring** expand **Additional configuration**.

- Uncheck **Enable Enhanced monitoring**.

28. Under **Additional configuration**, configure:

- **Initial database name:** `lab`
- Uncheck **Enable automatic backups**.
- Uncheck **Enable encryption**

This will turn off backups, which is not normally recommended, but will make the database deploy faster for this lab.

29. Choose **Create database**

Your database will now be launched.

If you receive an error that mentions "not authorized to perform: iam:CreateRole", make sure you unchecked *Enable Enhanced monitoring* in the previous step.

30. Choose **lab-db** (choose the link itself).

You will now need to wait **approximately 4 minutes** for the database to be available. The deployment process is deploying a database in two different Availability zones.

While you are waiting, you might want to review the [Amazon RDS FAQs](#) or grab a cup of coffee.

31. Wait until **Info** changes to **Modifying** or **Available**.

32. Scroll down to the **Connectivity & security** section and copy the **Endpoint** field.

It will look similar to: `lab-db.xxxx.us-east-1.rds.amazonaws.com`.

33. Paste the Endpoint value into a text editor. You will use it later in the lab.

Task 4: Interact with Your Database

In this task, you will open a web application running on a web server that has been created for you. You will configure it to use the database that you just created.

34. To discover the **WebServer** IP address, choose on the AWS Details drop down menu above these instructions. Copy the IP address value.

35. Open a new web browser tab, paste the *WebServer* IP address and press Enter.

The web application will be displayed, showing information about the EC2 instance.

36. Choose the **RDS** link at the top of the page.

You will now configure the application to connect to your database.

37. Configure the following settings:

- **Endpoint:** Paste the Endpoint you copied to a text editor earlier
- **Database:** lab
- **Username:** main
- **Password:** lab-password
- Choose **Submit**

A message will appear explaining that the application is running a command to copy information to the database. After a few seconds the application will display an **Address Book**.

The Address Book application is using the RDS database to store information.

38. Test the web application by adding, editing and removing contacts.

The data is being persisted to the database and is automatically replicating to the second Availability Zone.